

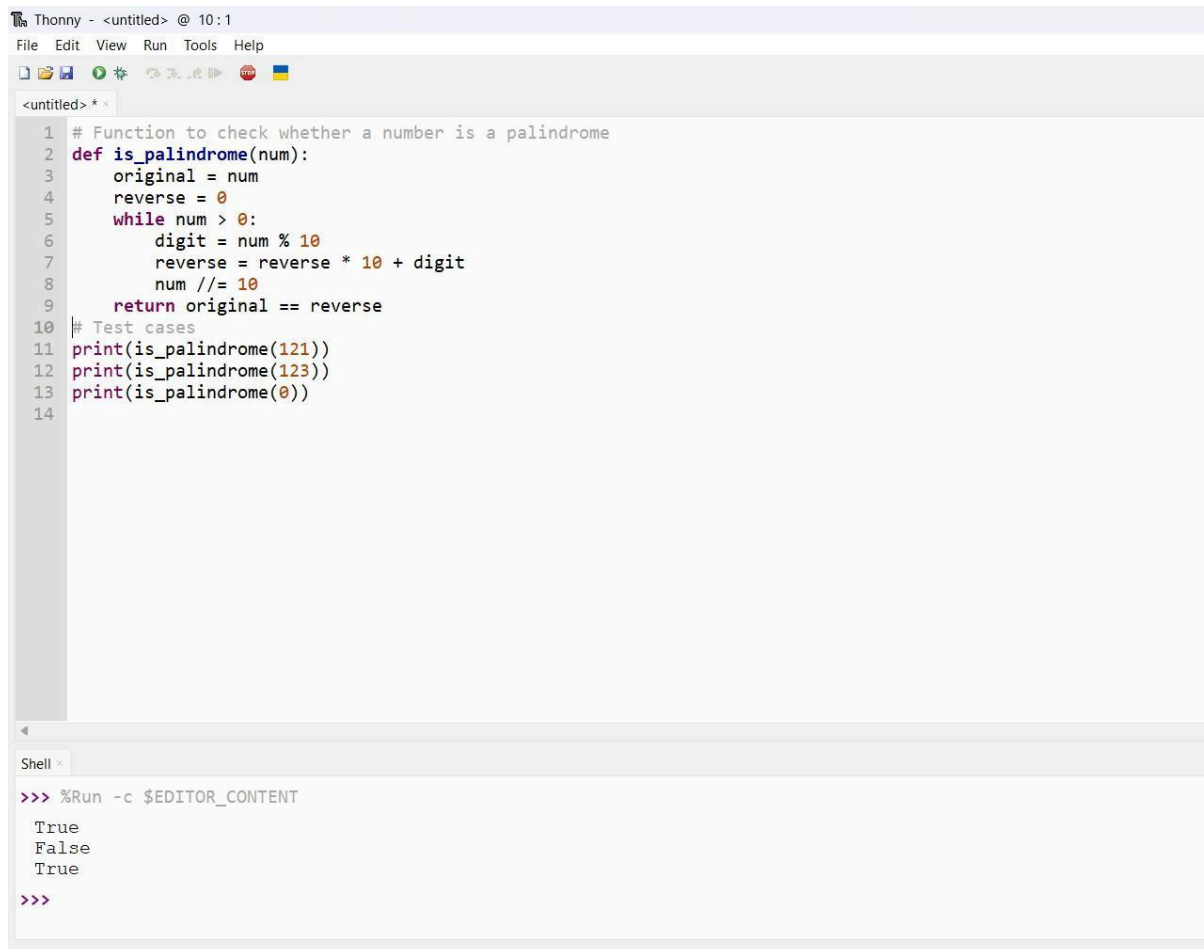
ASSIGNMENT-3

NAME:-J.SREEJA

BATCH-16

HTNO:-2303A51056

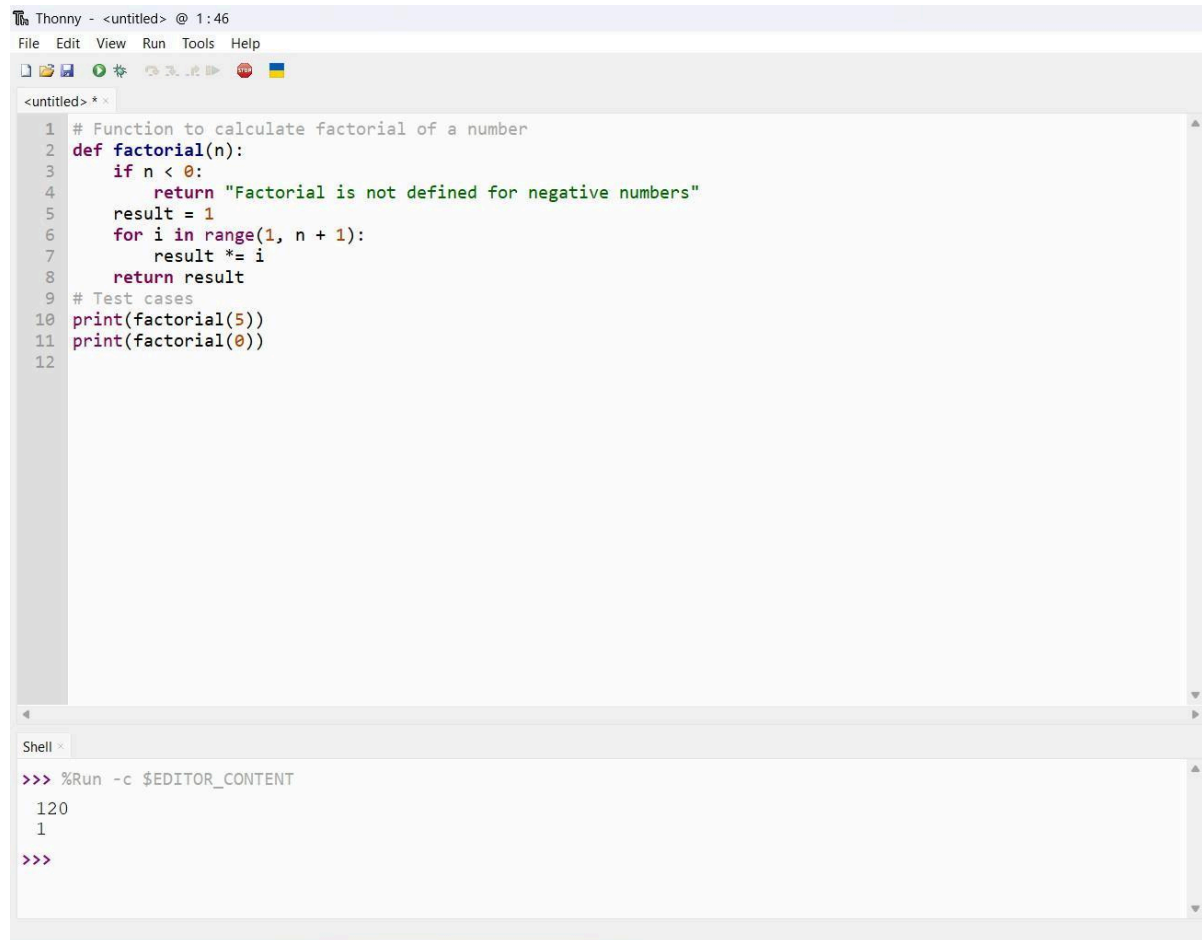
Zero-Shot Prompting – Palindrome Number Program:-



```
Thonny - <untitled> @ 10:1
File Edit View Run Tools Help
<untitled> * x
1 # Function to check whether a number is a palindrome
2 def is_palindrome(num):
3     original = num
4     reverse = 0
5     while num > 0:
6         digit = num % 10
7         reverse = reverse * 10 + digit
8         num //= 10
9     return original == reverse
10 # Test cases
11 print(is_palindrome(121))
12 print(is_palindrome(123))
13 print(is_palindrome(0))
14

Shell x
>>> %Run -c $EDITOR_CONTENT
True
False
True
>>>
```

One-Shot Prompting – Factorial Calculation:-



The image shows a screenshot of the Thonny Python IDE. The main editor window displays a Python script for calculating the factorial of a number. The script includes a function definition, a check for negative numbers, a loop for calculating the factorial, and test cases for 5 and 0. The bottom panel shows the output of the script, which is 120 for factorial(5) and 1 for factorial(0).

```
1 # Function to calculate factorial of a number
2 def factorial(n):
3     if n < 0:
4         return "Factorial is not defined for negative numbers"
5     result = 1
6     for i in range(1, n + 1):
7         result *= i
8     return result
9 # Test cases
10 print(factorial(5))
11 print(factorial(0))
12
```

Shell

```
>>> %Run -c $EDITOR_CONTENT
120
1
>>>
```

Few-Shot Prompting – Armstrong Number Check:-

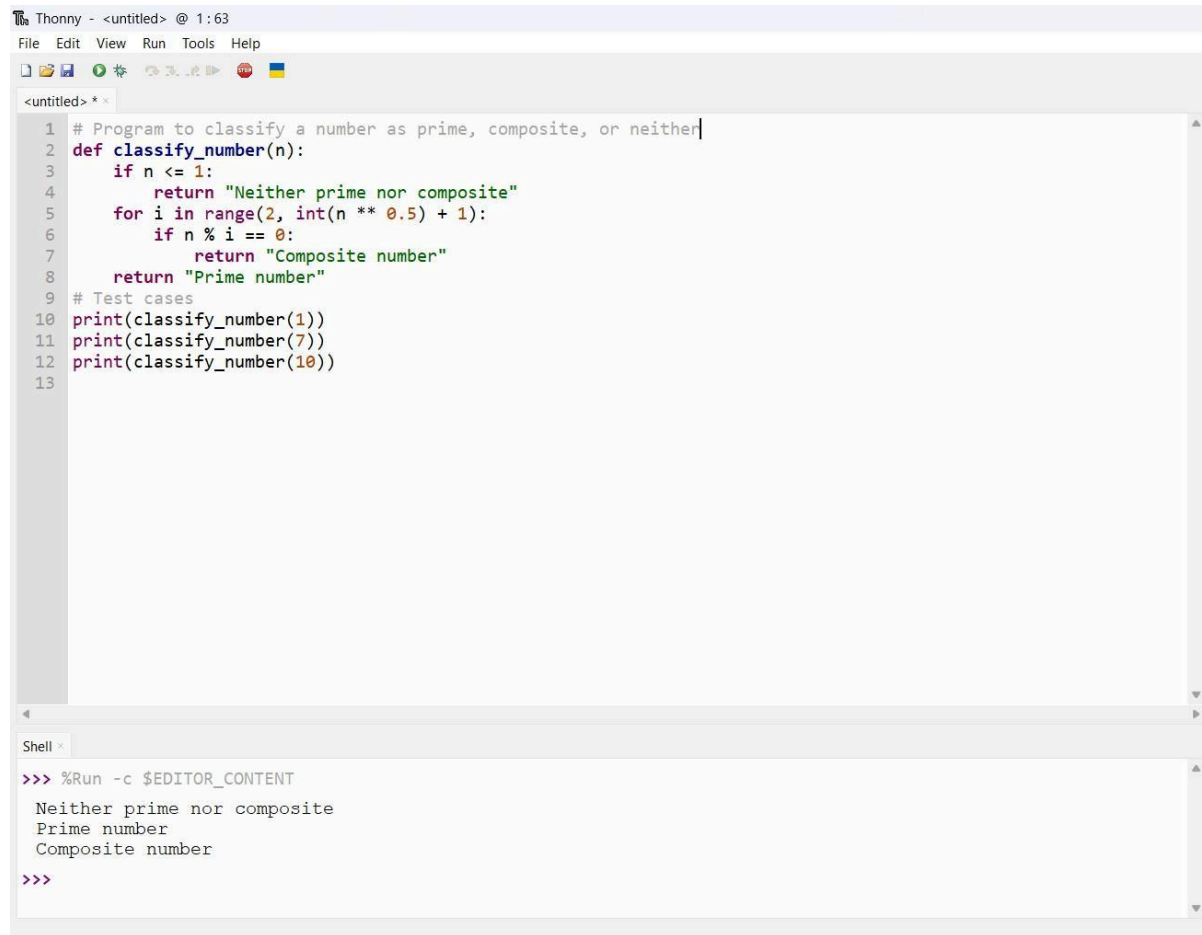
The image shows a screenshot of the Thonny Python IDE. The main editor window displays a Python script with the following code:

```
1 # Function to check whether a number is an Armstrong number
2 def is_armstrong(num):
3     digits = str(num)
4     power = len(digits)
5     total = 0
6     for digit in digits:
7         total += int(digit) ** power
8     return total == num
9 # Test cases
10 print(is_armstrong(153))
11 print(is_armstrong(370))
12 print(is_armstrong(123))
13
```

Below the editor, the Shell window shows the output of running the script:

```
>>> %Run -c $EDITOR_CONTENT
True
True
False
>>>
```

Question 4 (Optional): Context-Managed Prompting – Number Classification:-

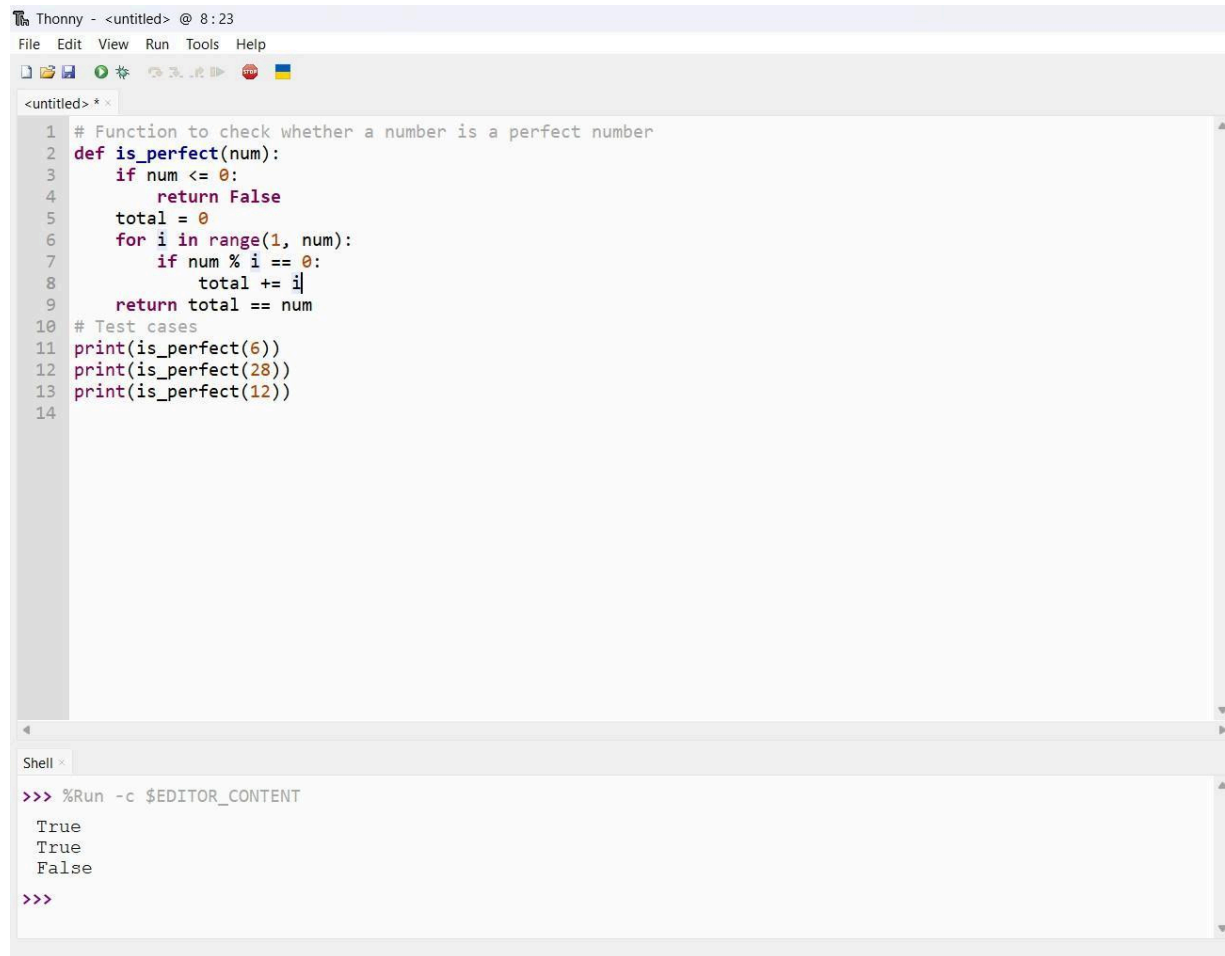


```
Thonny - <untitled> @ 1:63
File Edit View Run Tools Help

<untitled> * x
1 # Program to classify a number as prime, composite, or neither
2 def classify_number(n):
3     if n <= 1:
4         return "Neither prime nor composite"
5     for i in range(2, int(n ** 0.5) + 1):
6         if n % i == 0:
7             return "Composite number"
8     return "Prime number"
9 # Test cases
10 print(classify_number(1))
11 print(classify_number(7))
12 print(classify_number(10))
13

Shell x
>>> %Run -c $EDITOR_CONTENT
Neither prime nor composite
Prime number
Composite number
>>>
```

Question 5: Zero-Shot Prompting – Perfect Number Check:-



The image shows a screenshot of the Thonny Python IDE. The main editor window displays a Python script for checking perfect numbers. The script defines a function `is_perfect(num)` that returns `True` if the number is perfect and `False` otherwise. It includes test cases for 6, 28, and 12. Below the editor, the Shell window shows the output of running the script, which prints `True`, `True`, and `False` on separate lines.

```
1 # Function to check whether a number is a perfect number
2 def is_perfect(num):
3     if num <= 0:
4         return False
5     total = 0
6     for i in range(1, num):
7         if num % i == 0:
8             total += i
9     return total == num
10 # Test cases
11 print(is_perfect(6))
12 print(is_perfect(28))
13 print(is_perfect(12))
14
```

```
>>> %Run -c $EDITOR_CONTENT
True
True
False
>>>
```

Few-Shot Prompting – Even or Odd with Validation:-

Thonny - <untitled> @ 8:21

File Edit View Run Tools Help

<untitled> * x

```
1 # Program to check whether a number is even or odd with input validation
2 def even_or_odd(value):
3     if not isinstance(value, int):
4         return "Invalid input. Please enter an integer."
5     if value % 2 == 0:
6         return "Even"
7     else:
8         return "Odd"
9 # Test cases
10 print(even_or_odd(8))
11 print(even_or_odd(15))
12 print(even_or_odd(0))
13 print(even_or_odd(-7))
14
```

Shell x

```
>>> %Run -c $EDITOR_CONTENT
Even
Odd
Even
Odd
>>>
```